Year 13 Further Maths - Circular motion

Similar questions - Connected particles and the string through a hole

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Two particles A and B are connected by a light inextensible string of length 1.2 m. A has mass 1.5 kg and moves in a horizontal circle of radius 0.7 m on a smooth horizontal table, centre O. The string passes through a small smooth hole at O. B has mass 2 kg and moves in a horizontal circle below the table, with the string making an angle θ with the downward vertical through O, where $\cos\theta = 0.8$.

(a) Show that the length of string below the table is 0.5 m and find the tension in the string. **[4]**

(b) Determine the angular speeds of A and of B. **[5]**

(c) At one instant A, O and B lie in the same vertical plane with A and B on the same side of O. Find the time until they are next in the same vertical plane. **[3]**

2 A particle P of mass 0.4 kg moves in a circle of radius 0.6 m on a smooth horizontal table. It is attached to a light inextensible string passing through a smooth hole in the table, and a particle Q of mass 0.9 kg hangs in equilibrium at the other end.

(a) Find the speed of P. **[4]**

(b) Find the time for P to complete 10 revolutions. **[3]**

Question 1

(a) Length on the table is 0.7 m, so below the table is $1.2-0.7=0.5$ m. Vertically: $T\cos\theta = 2g$, $T = \frac{19.6}{0.8} = 24.5$ N

(b) For A: $T = 1.5\times 0.7\omega_A^2$, $\omega_A^2 = \frac{24.5}{1.05} = 23.33$, $\omega_A = 4.83\ \text{rad}\,\text{s}^{-1}$. For B (radius $0.5\sin\theta = 0.3$): $T\sin\theta = 2(0.3)\omega_B^2$ gives $\omega_B^2 = \frac{24.5\times 0.6}{0.6} = 24.5$, $\omega_B = 4.95\ \text{rad}\,\text{s}^{-1}$

(c) Relative angular speed 0.1192 rad s^{-1}; they are next in the same vertical plane (on opposite sides) after turning through a relative angle π: $t = \frac{\pi}{0.1192} = 26.4$ s

Question 2

(a) $T = 0.9g = 8.82$ N; $T = \frac{0.4v^2}{0.6}$, $v^2 = 13.23$, $v = 3.64\ \text{m}\,\text{s}^{-1}$

(b) One revolution takes $\frac{2\pi(0.6)}{3.637} = 1.037$ s so 10 revolutions take 10.4 s